

MINIMAL QUANTUM LOGIC: A BRIEF COMPARISON WITH CLASSICAL LOGIC

Ana Catarina Sousa

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MSc. in Mathematics
University of Minho

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SEMANTICAL CONTEXT

Boolean Algebras ^[3]

A structure $\mathcal{B} = (B; \vee, \wedge, ', 0, 1)$ is a **Boolean Algebra** if:

- $(B; \vee, \wedge)$ is a distributive lattice, ie:
For any $a, b, c \in B$: $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$;
- For any $a \in B$: $a \vee 0 = a$ and $a \wedge 1 = a$;
- For any $a \in B$: $a \vee a' = 1$ and $a \wedge a' = 0$.

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Example



Logical Truth

A formula A is a **logical truth** (or tautology) of Classical Logic (CL) if for any Boolean Algebra B , for any valuation v in B , $v(A) = 1$, where 1 is the maximum element.

Ortholattices ^[1]

A structure $\mathcal{B} = (B, \sqsubseteq, ', 1, 0)$ is an **ortholattice** if:

- $(B, \sqsubseteq)$ is a bounded lattice, with *maximum* **1** and *minimum* **0**;
- the unary operation $'$ (*orthocomplement*) satisfies the following conditions:
 - $a'' = a$;
 - if $a \sqsubseteq b$, then $b' \sqsubseteq a'$;
 - $a \sqcap a' = 0$.

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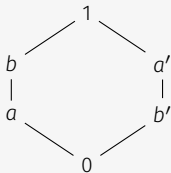


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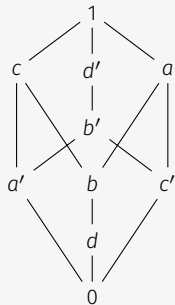


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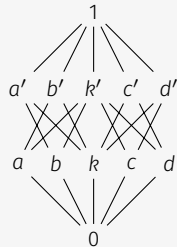


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Example



Orthomodular

For any $x, y \in B$:
if $x \sqsubseteq y$, then $y = x \sqcup (x' \sqcap y)$.

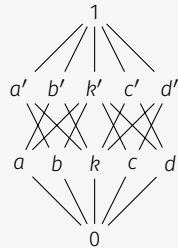
MINIMAL QUANTUM LOGIC AND THE ORTHOLATTICES

Ortholattices [1]

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Example



Orthomodular

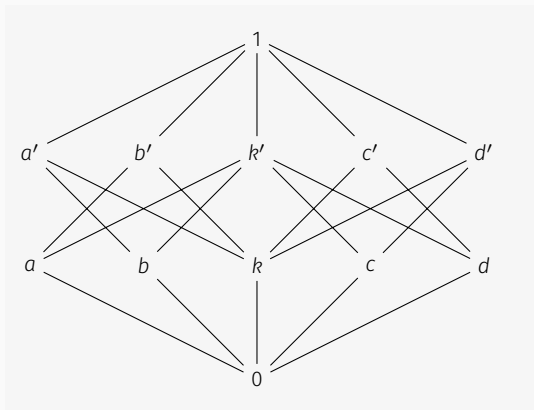
For any $x, y \in B$:
if $x \sqsubseteq y$, then $y = x \sqcup (x' \sqcap y)$.

Logical Truth

A formula A is a **logical truth** of Minimal Quantum Logic (MQL) if for any ortholattice B , for any valuation v in B , $v(A) = 1$, where **1** is the maximum element.

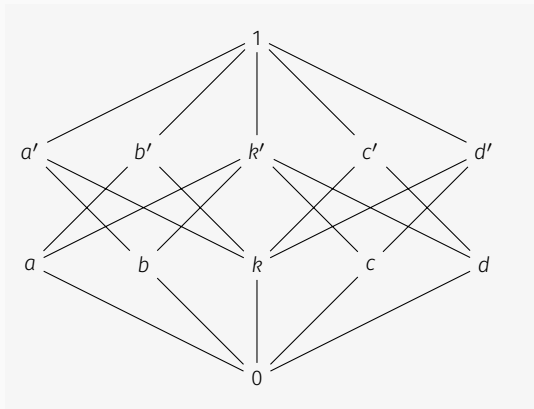
FAILURE OF DISTRIBUTIVE LAWS

In general, distributive laws don't hold for ortholattices.



FAILURE OF DISTRIBUTIVE LAWS

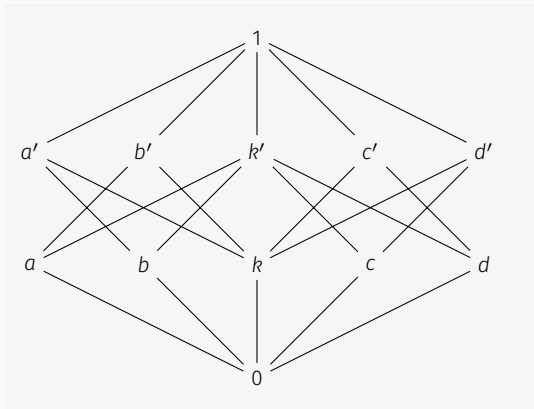
In general, distributive laws don't hold for ortholattices.



$$\begin{aligned} a \cap (b \sqcup c) &= a \\ &\neq \\ (a \cap b) \sqcup (a \cap c) &= 0 \end{aligned}$$

FAILURE OF DISTRIBUTIVE LAWS

In general, distributive laws don't hold for ortholattices.



$$\begin{aligned} a \sqcup (b \sqcap c) &= a \\ &\neq \\ (a \sqcup b) \sqcap (a \sqcup c) &= k' \end{aligned}$$

DEDUCTIVE SYSTEMS

AND THE CUT RULE RESTRICTIONS

Sequent

A **sequent** is an expression of the form

$$\Gamma \vdash \Theta$$

where Γ, Θ are finite (possibly empty) sets of formulas.

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Inference Rule

An **inference rule**

$$\frac{\Gamma_1 \vdash A_1, \dots, \Gamma_n \vdash A_n}{\Gamma \vdash A}$$

can be read as: if A_1 can be inferred from $\Gamma_1, \dots, A_n$ can be inferred from Γ_n , then A can be inferred from Γ .

INTRODUCTORY CONCEPTS

Sequent

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Deductive System

A **deductive system** is composed of a set of inference rules which chained together give rise to formal proofs, also called derivations.

Sequent Calculus for CL [5]

$$\frac{}{A \vdash A} \text{ id}$$

$$\frac{}{\perp \vdash A} \text{ L}\perp$$

$$\frac{\Gamma \vdash \Theta}{A, \Gamma \vdash \Theta} \text{ LW}$$

$$\frac{\Gamma \vdash \Theta}{\Gamma \vdash \Theta, A} \text{ RW}$$

$$\frac{\Gamma \vdash \Theta, A}{\neg A, \Gamma \vdash \Theta} \text{ L}\neg$$

$$\frac{A, \Gamma \vdash \Theta}{\Gamma \vdash \Theta, \neg A} \text{ R}\neg$$

$$\frac{A_i, \Gamma \vdash \Theta}{A_1 \wedge A_2, \Gamma \vdash \Theta} \text{ L}_i\wedge$$

$$\frac{\Gamma \vdash \Theta, A_1 \quad \Gamma \vdash \Theta, A_2}{\Gamma \vdash \Theta, A_1 \wedge A_2} \text{ R}\wedge$$

$$\frac{A_1, \Gamma \vdash \Theta \quad A_2, \Gamma \vdash \Theta}{A_1 \vee A_2, \Gamma \vdash \Theta} \text{ L}\vee$$

$$\frac{\Gamma \vdash \Theta, A_i}{\Gamma \vdash \Theta, A_1 \vee A_2} \text{ R}_i\vee$$

$$\frac{\Gamma \vdash \Theta, A \quad A, \Delta \vdash \Lambda}{\Gamma, \Delta \vdash \Theta, \Lambda} \text{ Cut}$$

SEQUENT CALCULUS

Sequent Calculus for CL [5]

$$\begin{array}{c}
 \frac{}{A \vdash A} \text{id} \qquad \frac{}{\perp \vdash A} L\perp \\
 \\
 \frac{\Gamma \vdash \Theta}{A, \Gamma \vdash \Theta} \text{LW} \qquad \frac{\Gamma \vdash \Theta}{\Gamma \vdash \Theta, A} \text{RW} \\
 \\
 \frac{\Gamma \vdash \Theta, A}{\neg A, \Gamma \vdash \Theta} L\neg \qquad \frac{A, \Gamma \vdash \Theta}{\Gamma \vdash \Theta, \neg A} R\neg \\
 \\
 \frac{A_i, \Gamma \vdash \Theta}{A_1 \wedge A_2, \Gamma \vdash \Theta} L_i\wedge \qquad \frac{\Gamma \vdash \Theta, A_1 \quad \Gamma \vdash \Theta, A_2}{\Gamma \vdash \Theta, A_1 \wedge A_2} R\wedge \\
 \\
 \frac{A_1, \Gamma \vdash \Theta \quad A_2, \Gamma \vdash \Theta}{A_1 \vee A_2, \Gamma \vdash \Theta} \text{LV} \qquad \frac{\Gamma \vdash \Theta, A_i}{\Gamma \vdash \Theta, A_1 \vee A_2} R_i\vee \\
 \\
 \frac{\Gamma \vdash \Theta, A \quad A, \Delta \vdash \Lambda}{\Gamma, \Delta \vdash \Theta, \Lambda} \text{Cut}
 \end{array}$$

Sequent Calculus for MQL [2, 4]

$$\begin{array}{c}
 \frac{}{A \vdash A} \text{id} \qquad \frac{\Gamma \vdash \Theta}{\Delta, \Gamma \vdash \Theta, \Lambda} W \\
 \\
 \frac{\Gamma \vdash \Theta}{\neg \Theta, \Gamma \vdash} L\neg \qquad \frac{\Gamma \vdash \Theta}{\Gamma \vdash \Theta, \neg \Gamma} R\neg \\
 \\
 \frac{A, \Gamma \vdash \Theta}{\neg \neg A, \Gamma \vdash \Theta} L\neg\neg \qquad \frac{\Gamma \vdash \Theta, A}{\Gamma \vdash \Theta, \neg \neg A} R\neg\neg \\
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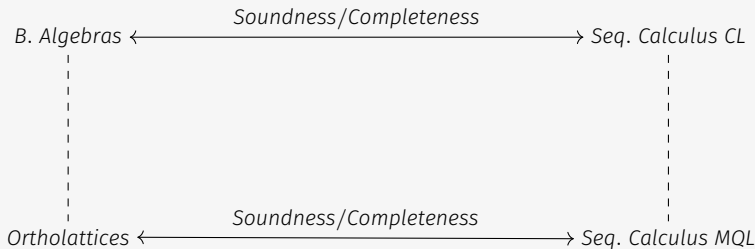
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 \\
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WHAT ARE THE CONSEQUENCES OF THESE RESTRICTIONS?

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First...

Semantical differences between Boolean algebras and Ortholattices are matched exactly by the differences between the sequent calculi for CL and MQL, because of the **Soundness** and **Completeness** meta-theorems enjoyed by both logics.



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The gap between the deductive systems lead to situations where it is more complex to derivate certain formulas in MQL, due to the imposed restrictions on the inference rules.

Finally...

There are some cases where it is even impossible to derivate a classical tautology in MQL.

CLASSICAL AND QUANTUM THEOREMS

WHOSE DERIVATIONS ARE MORE COMPLEX IN MQL

$$(A \vee B) \vee (\neg A \wedge \neg B)$$

$$(A \vee B) \vee (\neg A \wedge \neg B)$$

CL Derivation

$$\begin{array}{c}
 \frac{A \vdash A}{A \vdash A, B} \text{ RW} \qquad \frac{B \vdash B}{B \vdash A, B} \text{ RW} \\
 \frac{\vdash A, B, \neg A}{\vdash A, B, \neg A} \text{ R}\neg \qquad \frac{\vdash A, B, \neg B}{\vdash A, B, \neg B} \text{ R}\neg \\
 \hline
 \frac{\vdash A, B, \neg A \wedge \neg B}{\vdash A, B, \neg A \wedge \neg B} \text{ R}\wedge \\
 \hline
 \frac{\vdash A \vee B, \neg A \wedge \neg B}{\vdash A \vee B, \neg A \wedge \neg B} \text{ R}_1\vee, \text{ R}_2\vee \\
 \hline
 \vdash (A \vee B) \vee (\neg A \wedge \neg B) \text{ R}_1\vee, \text{ R}_2\vee
 \end{array}$$

$$(A \vee B) \vee (\neg A \wedge \neg B)$$

CL Derivation

$$\begin{array}{c}
 \frac{A \vdash A}{A \vdash A, B} \text{ RW} \quad \frac{B \vdash B}{B \vdash A, B} \text{ RW} \\
 \frac{\vdash A, B, \neg A}{\vdash A, B, \neg A} \text{ R}\neg \quad \frac{\vdash A, B, \neg B}{\vdash A, B, \neg B} \text{ R}\neg \\
 \hline
 \frac{\vdash A, B, \neg A \wedge \neg B}{\vdash A \vee B, \neg A \wedge \neg B} \text{ R}\wedge \\
 \hline
 \frac{\vdash A \vee B, \neg A \wedge \neg B}{\vdash (A \vee B) \vee (\neg A \wedge \neg B)} \text{ R}_1\vee, \text{ R}_2\vee
 \end{array}$$

CLASSICAL AND QUANTUM THEOREMS

$$(A \vee B) \vee (\neg A \wedge \neg B)$$

CL Derivation

$$\begin{array}{c}
 \frac{A \vdash A}{A \vdash A, B} \text{RW} \quad \frac{B \vdash B}{B \vdash A, B} \text{RW} \\
 \frac{\vdash A, B, \neg A}{\vdash A, B, \neg A} \text{R}\neg \quad \frac{\vdash A, B, \neg B}{\vdash A, B, \neg B} \text{R}\neg \\
 \hline
 \frac{\vdash A, B, \neg A \wedge \neg B}{\vdash A \vee B, \neg A \wedge \neg B} \text{R}\wedge \\
 \hline
 \frac{\vdash A \vee B, \neg A \wedge \neg B}{\vdash (A \vee B) \vee (\neg A \wedge \neg B)} \text{R}_1\vee, \text{R}_2\vee
 \end{array}$$

MQL Derivation

$$\begin{array}{c}
 \frac{A \vdash A}{A \vdash A, B} \text{W} \quad \frac{B \vdash B}{B \vdash A, B} \text{W} \\
 \frac{A \vdash A, B}{A \vdash A \vee B} \text{R}_1\vee, \text{R}_2\vee \quad \frac{B \vdash A, B}{B \vdash A \vee B} \text{R}_1\vee, \text{R}_2\vee \\
 \vdots \text{C*} \quad \vdots \text{C*} \\
 \hline
 \frac{A \vee B \vdash A \vee B}{\vdash A \vee B, \neg(A \vee B)} \text{id} \text{R}\neg \quad \frac{\neg(A \vee B) \vdash \neg A \quad \neg(A \vee B) \vdash \neg B}{\neg(A \vee B) \vdash \neg A \wedge \neg B} \text{R}\wedge \\
 \hline
 \frac{\vdash A \vee B, \neg(A \vee B) \quad \neg(A \vee B) \vdash \neg A \wedge \neg B}{\vdash A \vee B, \neg A \wedge \neg B} \text{Cut-1} \\
 \hline
 \frac{\vdash A \vee B, \neg A \wedge \neg B}{\vdash (A \vee B) \vee (\neg A \wedge \neg B)} \text{R}_1\vee, \text{R}_2\vee
 \end{array}$$

CLASSICAL AND QUANTUM THEOREMS

$$(A \vee B) \vee (\neg A \wedge \neg B)$$

CL Derivation	MQL Derivation
$ \begin{array}{c} \frac{A \vdash A}{A \vdash A, B} \text{ RW} \quad \frac{B \vdash B}{B \vdash A, B} \text{ RW} \\ \frac{\vdash A, B, \neg A}{\vdash A, B, \neg A} \text{ R}\neg \quad \frac{\vdash A, B, \neg B}{\vdash A, B, \neg B} \text{ R}\neg \\ \hline \frac{\vdash A, B, \neg A \wedge \neg B}{\vdash A \vee B, \neg A \wedge \neg B} \text{ R}\wedge \\ \hline \frac{\vdash A \vee B, \neg A \wedge \neg B}{\vdash (A \vee B) \vee (\neg A \wedge \neg B)} \text{ R}_1\vee, \text{ R}_2\vee \end{array} $	$ \begin{array}{c} \frac{A \vdash A}{A \vdash A, B} \text{ W} \quad \frac{B \vdash B}{B \vdash A, B} \text{ W} \\ \frac{A \vdash A, B}{A \vdash A \vee B} \text{ R}_1\vee, \text{ R}_2\vee \quad \frac{B \vdash A, B}{B \vdash A \vee B} \text{ R}_1\vee, \text{ R}_2\vee \\ \vdots \text{ C*} \quad \vdots \text{ C*} \\ \hline \frac{A \vee B \vdash A \vee B}{\vdash A \vee B, \neg(A \vee B)} \text{ id} \quad \frac{\neg(A \vee B) \vdash \neg A}{\neg(A \vee B) \vdash \neg A \wedge \neg B} \text{ R}\wedge \\ \hline \frac{\vdash A \vee B, \neg(A \vee B)}{\vdash A \vee B, \neg A \wedge \neg B} \text{ R}\neg \quad \frac{\neg(A \vee B) \vdash \neg A \wedge \neg B}{\vdash A \vee B, \neg A \wedge \neg B} \text{ Cut-1} \\ \hline \frac{\vdash A \vee B, \neg A \wedge \neg B}{\vdash (A \vee B) \vee (\neg A \wedge \neg B)} \text{ R}_1\vee, \text{ R}_2\vee \end{array} $

$(A \vee B) \vee (\neg A \wedge \neg B)$ is both a [CL theorem](#) and a [MQL theorem](#).

CLASSICAL THEOREMS

WHICH ARE NOT MQL THEOREMS

CLASSICAL (BUT NOT QUANTUM) THEOREMS

$$D := \neg(A \wedge (B \vee C)) \vee ((A \wedge B) \vee (A \wedge C))$$

CLASSICAL (BUT NOT QUANTUM) THEOREMS

$$D := \neg(A \wedge (B \vee C)) \vee ((A \wedge B) \vee (A \wedge C))$$

CL Derivation

$$\begin{array}{c}
 \frac{\frac{A \vdash A}{A, B \vdash A, A \wedge C} \text{ LW, RW} \quad \frac{\frac{B \vdash B}{A, B \vdash B, A \wedge C} \text{ LW, RW}}{A, B \vdash A \wedge B, A \wedge C} \text{ R}\wedge \quad \frac{\frac{A \vdash A}{A, C \vdash A \wedge B, A} \text{ LW, RW} \quad \frac{\frac{C \vdash C}{A, C \vdash A \wedge B, C} \text{ LW, RW}}{A, C \vdash A \wedge B, A \wedge C} \text{ R}\wedge \\
 \frac{A, B \vdash A \wedge B, A \wedge C \quad A, C \vdash A \wedge B, A \wedge C}{A, B \vee C \vdash A \wedge B, A \wedge C} \text{ L}\vee \\
 \frac{A, B \vee C \vdash A \wedge B, A \wedge C}{A, B \vee C \vdash (A \wedge B) \vee (A \wedge C)} \text{ R}_1\vee, \text{ R}_2\vee \\
 \frac{A, B \vee C \vdash (A \wedge B) \vee (A \wedge C)}{A \wedge (B \vee C) \vdash (A \wedge B) \vee (A \wedge C)} \text{ L}_1\wedge, \text{ L}_2\wedge \\
 \frac{A \wedge (B \vee C) \vdash (A \wedge B) \vee (A \wedge C)}{\vdash \neg(A \wedge (B \vee C)), (A \wedge B) \vee (A \wedge C)} \text{ R}\neg \\
 \frac{\vdash \neg(A \wedge (B \vee C)), (A \wedge B) \vee (A \wedge C)}{\vdash \neg(A \wedge (B \vee C)) \vee ((A \wedge B) \vee (A \wedge C))} \text{ R}_1\vee, \text{ R}_2\vee
 \end{array}$$

CLASSICAL (BUT NOT QUANTUM) THEOREMS

$$D := \neg(A \wedge (B \vee C)) \vee ((A \wedge B) \vee (A \wedge C))$$

CL Derivation

$$\begin{array}{c}
 \frac{\frac{A \vdash A}{A, B \vdash A, A \wedge C} \text{ LW, RW} \quad \frac{\frac{B \vdash B}{A, B \vdash B, A \wedge C} \text{ LW, RW}}{A, B \vdash A \wedge B, A \wedge C} \text{ R}\wedge \quad \frac{\frac{\frac{A \vdash A}{A, C \vdash A \wedge B, A} \text{ LW, RW} \quad \frac{\frac{C \vdash C}{A, C \vdash A \wedge B, C} \text{ LW, RW}}{A, C \vdash A \wedge B, A \wedge C} \text{ R}\wedge} \\
 \frac{A, B \vdash A \wedge B, A \wedge C \quad A, C \vdash A \wedge B, A \wedge C}{A, B \vee C \vdash A \wedge B, A \wedge C} \text{ L}\vee \\
 \frac{A, B \vee C \vdash A \wedge B, A \wedge C}{A, B \vee C \vdash (A \wedge B) \vee (A \wedge C)} \text{ R}_1\vee, \text{ R}_2\vee \\
 \frac{A, B \vee C \vdash (A \wedge B) \vee (A \wedge C)}{A \wedge (B \vee C) \vdash (A \wedge B) \vee (A \wedge C)} \text{ L}_1\wedge, \text{ L}_2\wedge \\
 \frac{A \wedge (B \vee C) \vdash (A \wedge B) \vee (A \wedge C)}{\vdash \neg(A \wedge (B \vee C)), (A \wedge B) \vee (A \wedge C)} \text{ R}\neg \\
 \frac{\vdash \neg(A \wedge (B \vee C)), (A \wedge B) \vee (A \wedge C)}{\vdash \neg(A \wedge (B \vee C)) \vee ((A \wedge B) \vee (A \wedge C))} \text{ R}_1\vee, \text{ R}_2\vee
 \end{array}$$

D is a CL theorem.

CLASSICAL (BUT NOT QUANTUM) THEOREMS

$$D := \neg(A \wedge (B \vee C)) \vee ((A \wedge B) \vee (A \wedge C))$$

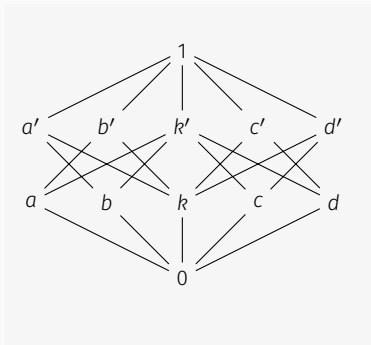
CL Derivation

$$\begin{array}{c}
 \frac{\frac{A \vdash A}{A, B \vdash A, A \wedge C} \text{ LW, RW} \quad \frac{\frac{B \vdash B}{A, B \vdash B, A \wedge C} \text{ LW, RW}}{A, B \vdash A \wedge B, A \wedge C} \text{ R}\wedge \quad \frac{\frac{\frac{A \vdash A}{A, C \vdash A \wedge B, A} \text{ LW, RW} \quad \frac{\frac{C \vdash C}{A, C \vdash A \wedge B, C} \text{ LW, RW}}{A, C \vdash A \wedge B, A \wedge C} \text{ R}\wedge} \\
 \frac{A, B \vdash A \wedge B, A \wedge C \quad A, C \vdash A \wedge B, A \wedge C}{A, B \vee C \vdash A \wedge B, A \wedge C} \text{ L}\vee \\
 \frac{A, B \vee C \vdash A \wedge B, A \wedge C}{A, B \vee C \vdash (A \wedge B) \vee (A \wedge C)} \text{ R}_1\vee, \text{ R}_2\vee \\
 \frac{A \wedge (B \vee C) \vdash (A \wedge B) \vee (A \wedge C)}{\vdash \neg(A \wedge (B \vee C)), (A \wedge B) \vee (A \wedge C)} \text{ L}_1\wedge, \text{ L}_2\wedge \\
 \frac{\vdash \neg(A \wedge (B \vee C)), (A \wedge B) \vee (A \wedge C)}{\vdash \neg(A \wedge (B \vee C)) \vee ((A \wedge B) \vee (A \wedge C))} \text{ R}\neg \\
 \frac{\vdash \neg(A \wedge (B \vee C)) \vee ((A \wedge B) \vee (A \wedge C))}{\vdash \neg(A \wedge (B \vee C)) \vee ((A \wedge B) \vee (A \wedge C))} \text{ R}_1\vee, \text{ R}_2\vee
 \end{array}$$

D is a CL theorem.

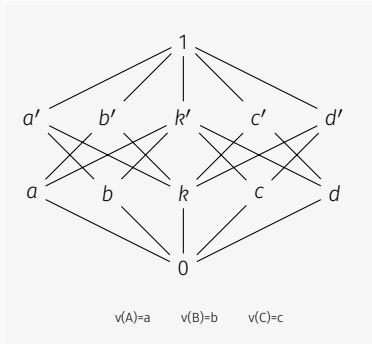
CLASSICAL (BUT NOT QUANTUM) THEOREMS

$$D := \neg(A \wedge (B \vee C)) \vee ((A \wedge B) \vee (A \wedge C))$$



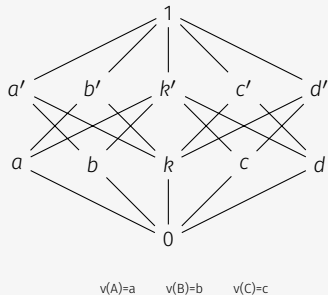
CLASSICAL (BUT NOT QUANTUM) THEOREMS

$$D := \neg(A \wedge (B \vee C)) \vee ((A \wedge B) \vee (A \wedge C))$$



CLASSICAL (BUT NOT QUANTUM) THEOREMS

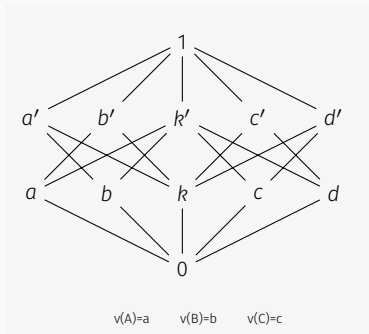
$$D := \neg(A \wedge (B \vee C)) \vee ((A \wedge B) \vee (A \wedge C))$$



$$\begin{aligned} v(D) &= v(\neg(A \wedge (B \vee C)) \vee ((A \wedge B) \vee (A \wedge C))) \\ &= \dots \\ &= ((a \sqcap (b' \sqcap c'))') \sqcap ((a \sqcap b)' \sqcap (a \sqcap c'))' \\ &= ((a \sqcap k') \sqcap (0' \sqcap 0'))' \\ &= (a \sqcap 1)' \\ &= a' \\ &\neq 1. \end{aligned}$$

CLASSICAL (BUT NOT QUANTUM) THEOREMS

$$D := \neg(A \wedge (B \vee C)) \vee ((A \wedge B) \vee (A \wedge C))$$

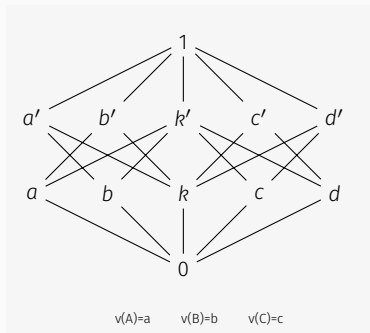


$$\begin{aligned} v(D) &= v(\neg(A \wedge (B \vee C)) \vee ((A \wedge B) \vee (A \wedge C))) \\ &= \dots \\ &= ((a \sqcap (b' \sqcap c')') \sqcap ((a \sqcap b)' \sqcap (a \sqcap c)'))' \\ &= ((a \sqcap k') \sqcap (0' \sqcap 0'))' \\ &= (a \sqcap 1)' \\ &= a' \\ &\neq 1. \end{aligned}$$

D is not a ML logical truth.

CLASSICAL (BUT NOT QUANTUM) THEOREMS

$$D := \neg(A \wedge (B \vee C)) \vee ((A \wedge B) \vee (A \wedge C))$$



$$\begin{aligned} v(D) &= v(\neg(A \wedge (B \vee C)) \vee ((A \wedge B) \vee (A \wedge C))) \\ &= \dots \\ &= ((a \sqcap (b' \sqcap c'))' \sqcap ((a \sqcap b)' \sqcap (a \sqcap c')))' \\ &= ((a \sqcap k') \sqcap (0' \sqcap 0'))' \\ &= (a \sqcap 1)' \\ &= a' \\ &\neq 1. \end{aligned}$$

Hence, by the soundness meta-theorem, D is not a MQT theorem.



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